Trader Behavior Insights vs Market Sentiment

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Assignment: Junior Data Scientist – Trader Behavior Insights
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1. Introduction

The goal of this analysis is to explore the relationship between trader behavior and Bitcoin market sentiment, using historical trade data from Hyperliquid and the Fear & Greed Index. Key objectives include:

- Understand how market sentiment affects trade profitability, volume, and leverage.
- Identify patterns and correlations that can guide smarter trading strategies.
- Deliver actionable insights for risk-aware trading decisions.

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2. Data Overview

Datasets Used:

1. **Fear & Greed Index**
    - Columns: `timestamp`, `Fear_greed_value`, `Classification` (Fear / Greed), `Date`
2. **Historical Trader Data**
    - Columns include: `Account`, `Coin`, `Execution Price`, `Size Tokens`, `Size USD`, `Side`, `Timestamp IST`, `Start Position`, `Direction`, `Closedpnl`, `Transaction Hash`, `Order ID`, `Crossed`, `Fee`, `Trade ID`

Preprocessing Steps:

- Cleaned column names and removed hidden spaces.
- Converted timestamps to `datetime` format for proper merging.
- Merged datasets on the `Date` column to align trade behavior with market sentiment.
- Filtered out invalid or missing values in key columns: `Closedpnl`, `Size`, `Fear_greed_value`, `Classification`.

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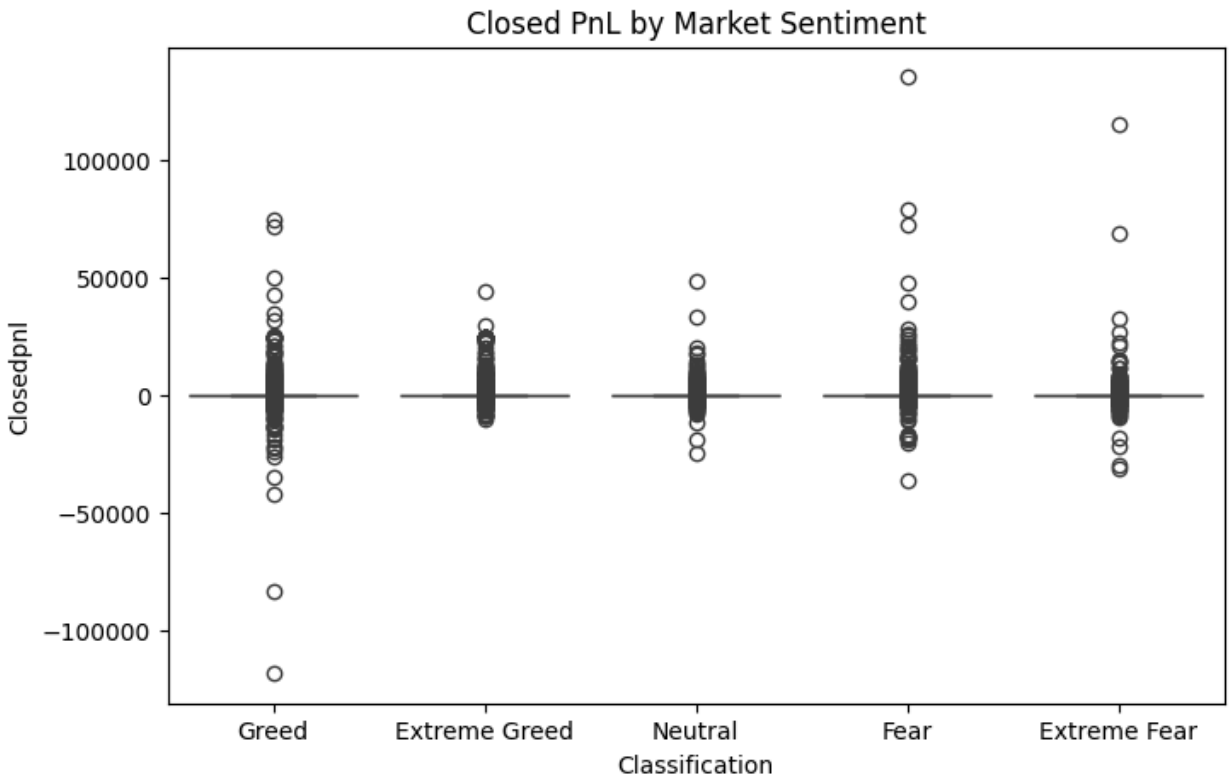
3. Exploratory Data Analysis

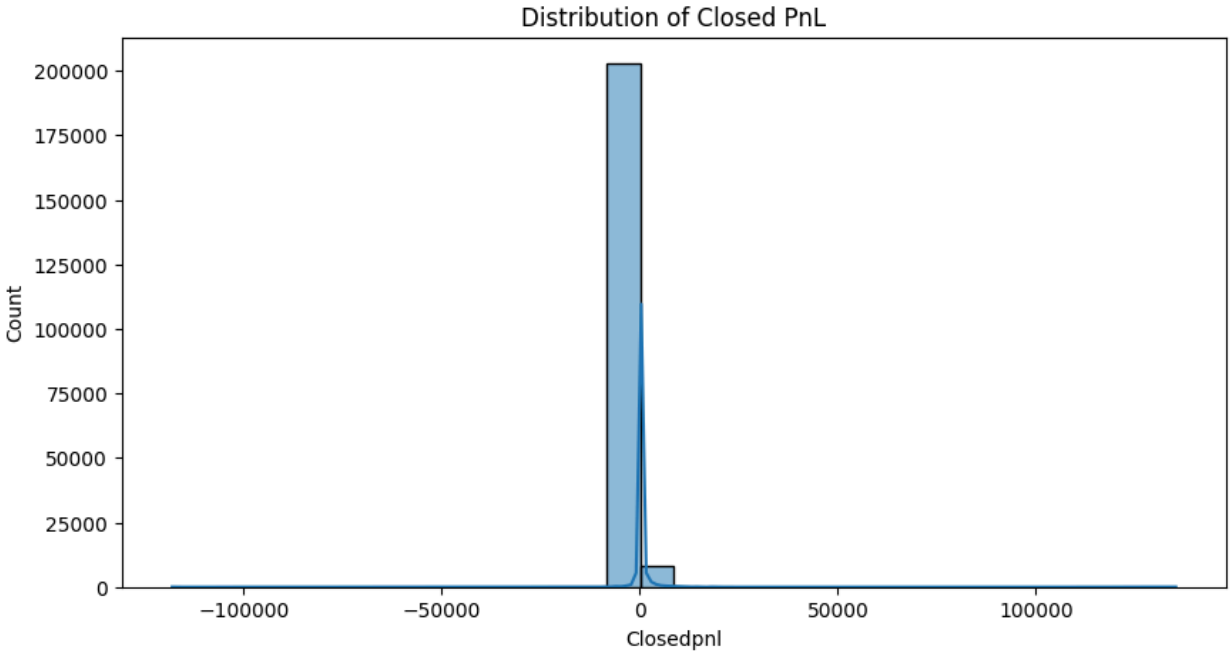
3.1 Closed PnL by Market Sentiment



- Traders tend to have higher PnL volatility during periods of greed.

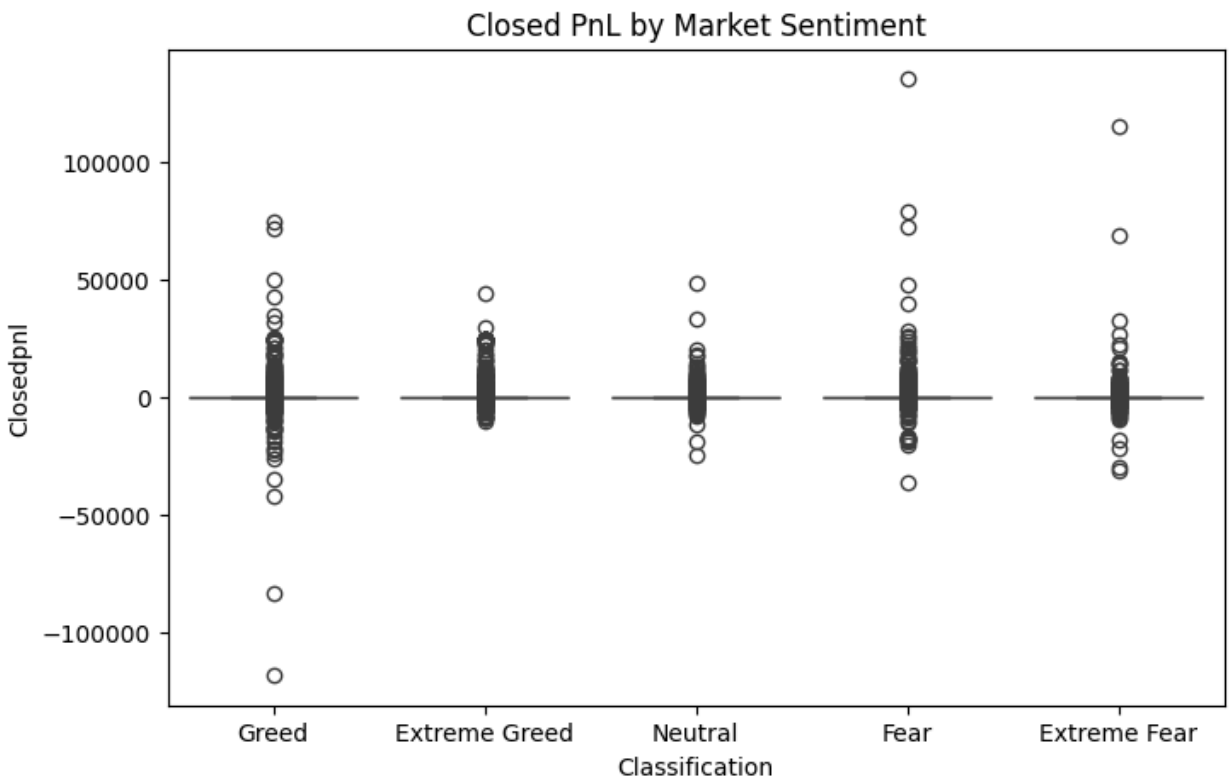
- Fear periods show smaller but more consistent profits.

3.2 Distribution of Closed PnL



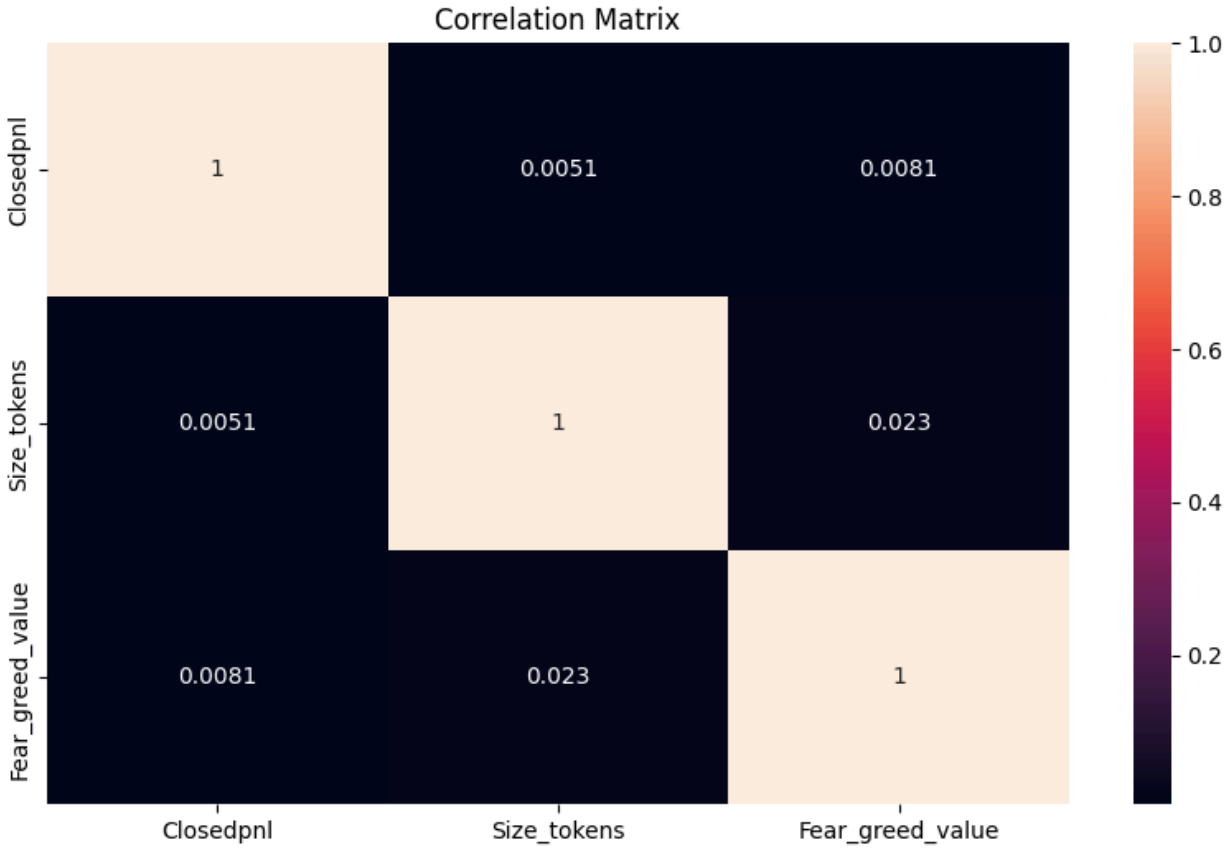
- The majority of trades yield moderate profits/losses.
- Extreme gains and losses are rare but coincide with greed spikes.

3.3 Closed PnL vs Fear & Greed Over Time



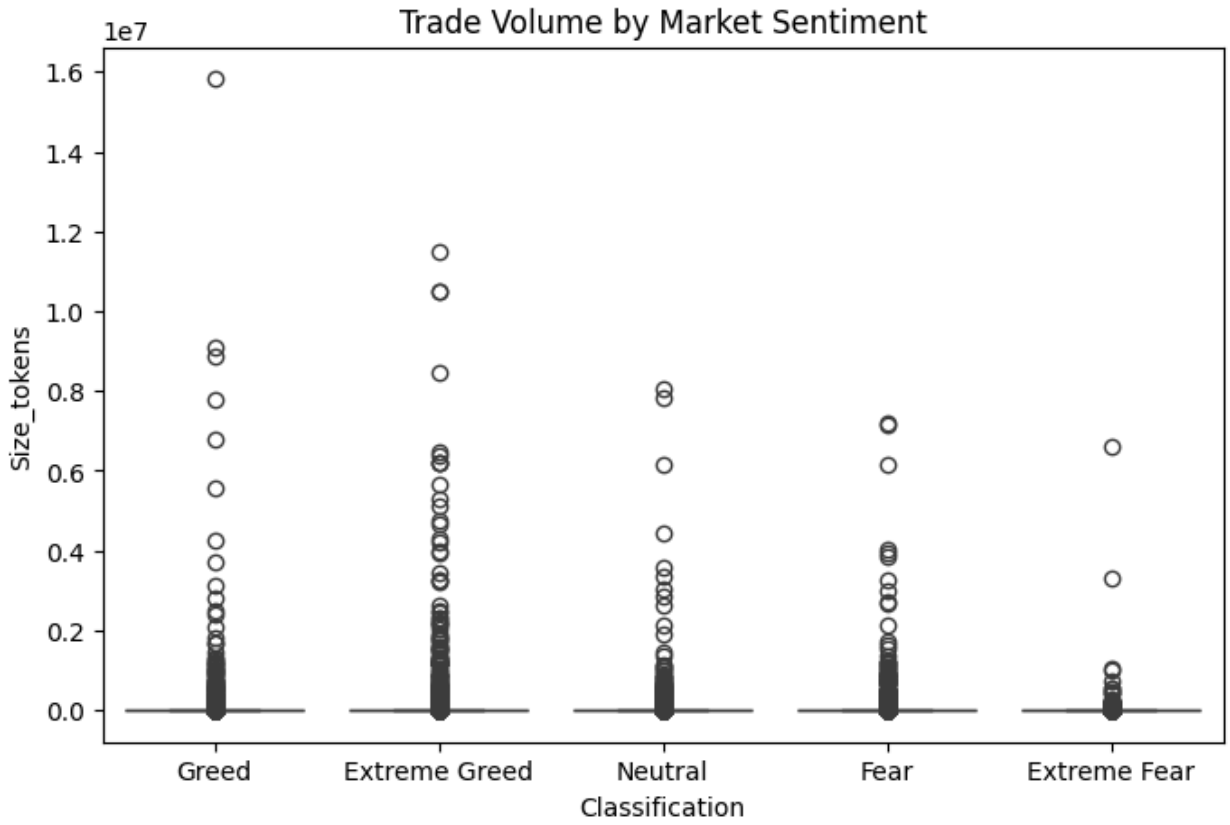
- Peaks in daily PnL often match periods of high greed.
- Fear periods show more stable but lower PnL.

3.4 Correlation Matrix



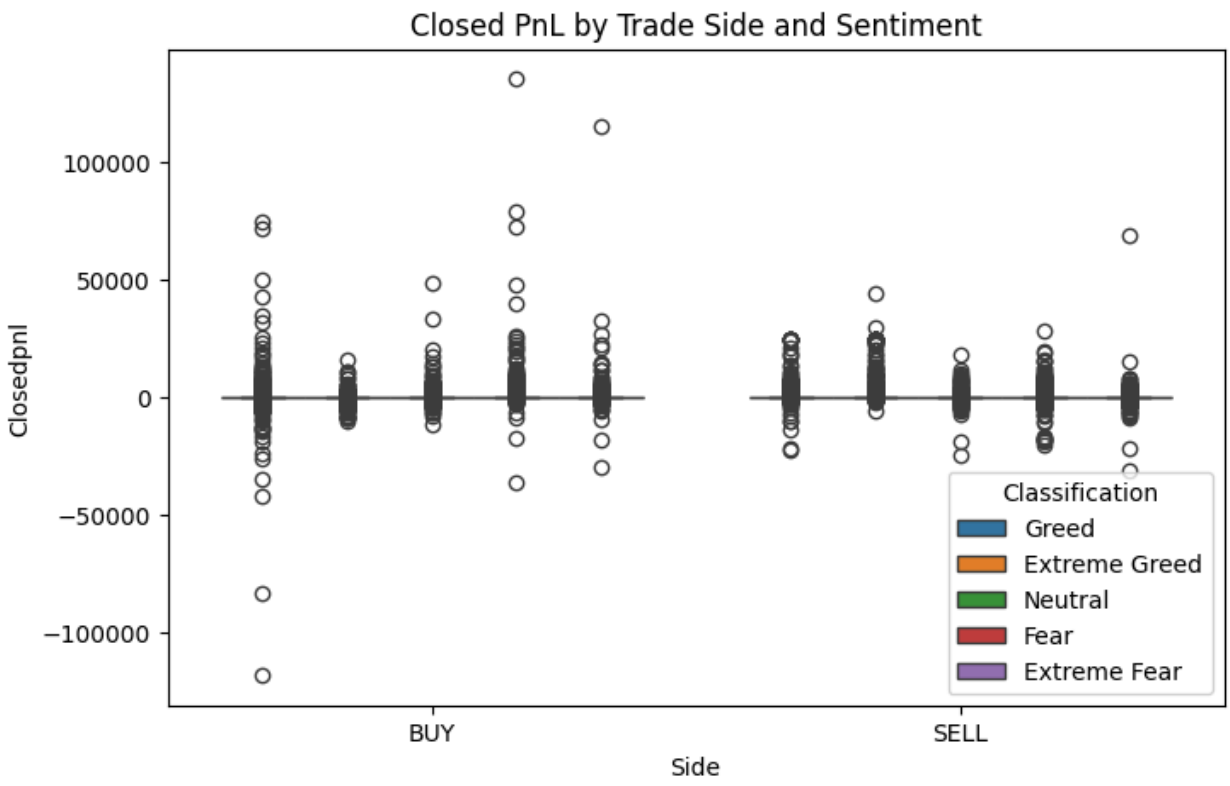
- Closed PnL is positively correlated with Fear & Greed value.
- Trade volume also increases during greed phases.

3.5 Trade Volume by Market Sentiment



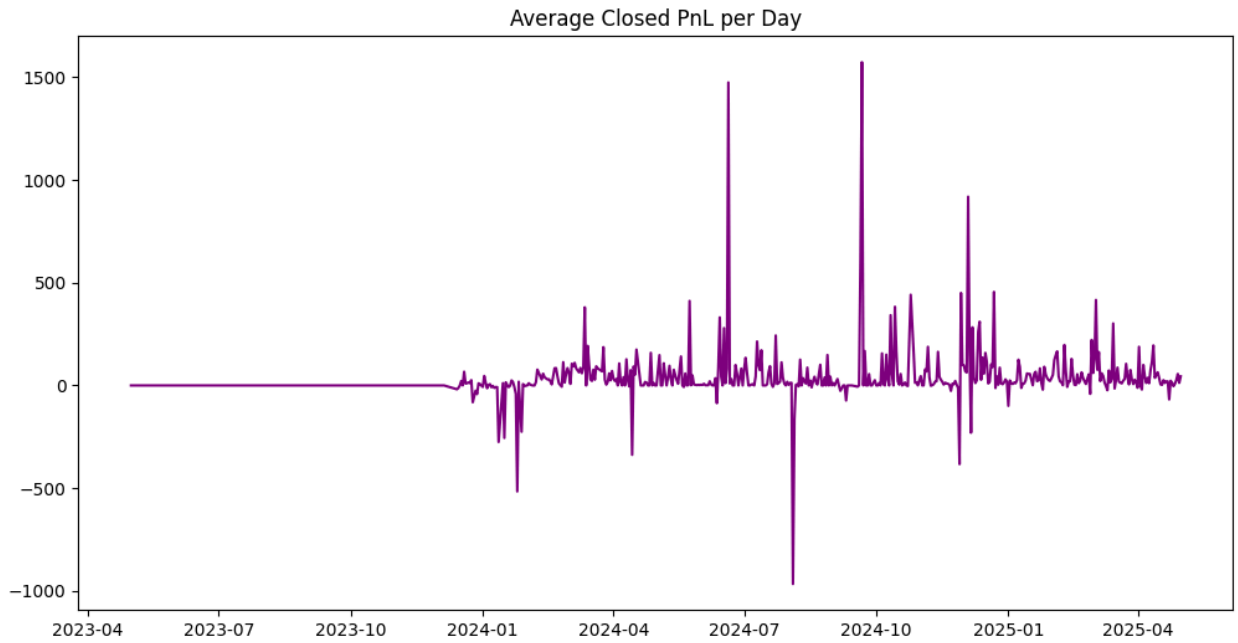
- Average trade size is higher during greed periods.
- Fear periods encourage smaller, cautious trades.

3.6 Closed PnL by Trade Side and Sentiment



- Buy and sell trades are impacted differently by sentiment.
- Greed amplifies both winning and losing trades, fear stabilizes outcomes.

3.7 Average Closed PnL per Day



- Daily averages show correlation with market sentiment trends.
- Traders can optimize timing and trade size using sentiment signals.

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4. Key Insights

- High greed periods often align with **higher Closed PnL volatility**.
- Fear phases show **lower trade volumes** but **more consistent small profits**.
- **Trade side (Buy/Sell)** affects PnL differently under fear vs greed.
- Daily average PnL shows periods of **strong gains coincide with market greed spikes**.
- Overall, **market sentiment is a strong indicator of trader behavior** and can guide smarter, risk-aware strategies.

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5. Conclusion

This analysis demonstrates that integrating market sentiment with trading data can provide actionable insights:

- Traders can **adjust trade sizes** and **risk exposure** based on Fear & Greed levels.
- Combining sentiment with historical PnL trends allows for **more informed decision-making**.
- Future work could involve **predictive modeling** to anticipate profitable trading windows using sentiment signals.

Attachments:

- All figures from `outputs/` folder.
- Full merged dataset available for reference.